

Year 12 Pre-Mock Recap — cumulative 1.1–1.4 + 2.1 + 2.2.1

Name: _____ Date: _____ Mark: _____ / 60

Time allowed: 75 minutes. Closed book.

OCR H446 cumulative pre-mock recap. Section A covers Component 1 (36 marks); Section B covers Component 2 (24 marks). Show all working.

Questions

Section A — Component 1 (36 marks)

Q1. Describe the role of the **address bus**, the **data bus** and the **control bus** when the CPU fetches an instruction from main memory. **[3]** — 1.1.1

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Q2. A desktop computer contains both **RAM** and **ROM**. (a) State **two** differences between RAM and ROM. **[2]**

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(b) Give **one** reason the computer needs ROM. **[1]**

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[3 total] — 1.1.3

Q3. An operating system is managing three processes. (a) Describe how **round robin** scheduling shares the processor between the processes. **[2]**

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(b) Explain how a long job could be treated unfairly under **shortest job first** scheduling. [1]

(c) State the name of the memory-management technique that divides memory into **fixed-size** blocks. [1]

[4 total] — 1.2.1

Q4. A developer writing firmware for a microcontroller programs in **assembly language**. (a) State the name of the translator that converts assembly language into machine code. [1]

(b) Give **one** reason assembly language is appropriate for this task. [1]

(c) Within a machine-code instruction, state what is meant by the **opcode** and the **operand**. [1]

[3 total] — 1.2.4

Q5. A student types `www.revisionhub.co.uk` into a web browser. (a) Describe how **DNS** is used to obtain the IP address of the web server, including what happens when the local resolver does **not** already know the address. [3]

(b) State what is meant by a **protocol**, and give **one** reason standard protocols are essential on the internet. [2]

[5 total] — 1.3.3

Q6. Show all working. (a) Convert the denary number **154** into 8-bit binary. [1]

(b) Convert **154** into hexadecimal. [1]

(c) Using **8-bit two's complement**, calculate **47 – 61**: show 47 and 61 in 8-bit binary, form the two's complement of 61, then perform the addition and state the final **denary** result. [3]

[5 total] — 1.4.1

Q7. Show each step. (a) Use **De Morgan's law** to rewrite $\neg(B \vee \neg C)$ so that NOT is no longer applied to a bracket. [2]

(b) Simplify $A \wedge B \vee A \wedge B \wedge C$, naming the law used. [2]

[4 total] — 1.4.3

Q8. A sorted collection of values is stored in a **linked list**: `14 → 26 → 33`. (a) Describe how a linked list stores its items, referring to **nodes** and **pointers**. [2]

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(b) Describe the steps needed to insert the value **29** into its correct position, and state why no existing data has to be moved. [2]

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[4 total] — 1.4.2

Q9. A music festival issues electronic wristbands. Battery-powered scanners at each gate currently check every wristband against a **central database** over the site's Wi-Fi; the organisers are considering instead giving each scanner its **own local copy** of the ticket data. Evaluate the two approaches, referring to **data consistency**, **reliability of the network link**, and the **security of transmitted data**. [5] — *synoptic 1.3.1 / 1.3.2 / 1.3.3*

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Section B — Component 2 (24 marks)

Q10. A city introduces a bike-hire scheme: riders unlock a bike at a docking station using an app, ride it, and return it to any station. The app shows a map of stations with the number of bikes available at each.

(a) Identify **one** example of **abstraction** in the app's map. [1]

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(b) The developers **decompose** the overall problem. Identify **two** sub-problems of "hire a bike". [2]

(c) Identify **one** part of the system that is well suited to **concurrent** processing. [1]

[4 total] — 2.1.1 / 2.1.3 / 2.1.5

Q11. A toll-bridge barrier rises only when payment has been accepted **and** the vehicle's height, held in the variable `height`, does not exceed **2.4** metres. The Boolean variable `paymentOK` is available. (a) *Thinking logically*: write the **Boolean condition** the program should test before raising the barrier. [2]

(b) *Thinking ahead*: the toll charge looked up for each regular number plate could be **cached** after its first lookup. State **one benefit** and **one risk** of this caching. [2]

[4 total] — 2.1.4 / 2.1.2

Q12. The following program is written in OCR Exam Reference Language.

```
n = 305
rev = 0
while n > 0
    rev = rev * 10 + (n MOD 10)
    n = n DIV 10
endwhile
print(rev)
```

(a) Complete the trace table, showing the values of `n` and `rev` at the **end of each iteration** of the loop. [3]

Iteration	n	rev
start	305	0
1		
2		
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(b) State the value output by the program. [1]

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(c) State the purpose of the program. [1]

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[5 total] — 2.2.1

Q13. A swimming coach records a swimmer's lap times, in seconds (real numbers), in the array `laps`. Write a function `countFaster(laps, target)` in **OCR Exam Reference Language** that: - returns `-1` if `laps` is empty; - otherwise returns the **number of laps** swum in **strictly less than** `target` seconds. Use `laps.length` to find the number of items in the array. [6] — 2.2.1

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Q14. A programmer is improving a menu-driven program. (a) State **two** features of an **IDE** that help the programmer find the cause of a logic error, describing how each helps. [2]

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(b) Give **one** reason for using a **named constant** rather than repeating a literal value through the code. [1]

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(c) The menu must be displayed **at least once**, and redisplayed until the user enters a valid choice. Name the most suitable **iteration structure** in OCR Exam Reference Language and justify your choice over `while ... endwhile`. [2]

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[5 total] — 2.2.1
